

Supplementary Appendix: Graph-Based and Self-Supervised Knowledge Tracing in Sparse-Concept Settings

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1 OVERVIEW OF SUPPLEMENTARY ARTIFACTS

This supplementary appendix accompanies the paper *"Graph-Based and Self-Supervised Knowledge Tracing in Sparse-Concept Settings: A Systematic Survey and Research Agenda"*. It provides full methodological traceability, database search logs, quality assessment breakdowns (QA1–QA8), and full coding logs across all 42 primary core studies.

2 DATABASE SEARCH LOGS & DEDUPLICATION MATRIX

Search queries were executed across 7 electronic databases for the modern deep learning era (2015-01-01 to 2026-07-31):

- **Scopus:** 531 records
- **Web of Science Core Collection:** 344 records
- **ScienceDirect:** 257 records
- **SpringerLink:** 255 records
- **IEEE Xplore:** 232 records
- **ACM Digital Library:** 200 records
- **arXiv Preprints:** 198 records

Total Raw Records: 1,935. Automated deduplication removed 1,185 records, yielding 750 unique deduplicated records. Title/Abstract screening excluded 610 records. Full-text eligibility assessment of 140 articles resulted in 98 exclusions (EC1–EC7), leaving an integrated core corpus of **42 primary core studies**.

3 QUALITY ASSESSMENT (QA1–QA8) SUMMARY MATRIX

Mean Quality Score across 42 primary core studies: **0.7768** (MODERATE-HIGH Quality).

- **QA1 Task Clarity:** 1.95 / 2.00
- **QA2 Data Transparency:** 1.90 / 2.00
- **QA3 Split Control:** 1.76 / 2.00
- **QA4 Baseline Adequacy:** 1.90 / 2.00
- **QA5 Statistical Rigor:** 1.10 / 2.00
- **QA6 Reproducibility Artifacts:** 1.45 / 2.00
- **QA7 Sparse Construct Validity:** 1.05 / 2.00
- **QA8 Computational Transparency:** 1.38 / 2.00

4 PRIMARY CORE STUDY INDEX (42 CORE PAPERS)

The primary core corpus comprises 42 studies published between 2015 and 2026, including landmark GNN and SSL models such as GKT [?], GIKT [?], Bi-CLKT [?], DyGKT [?], CL4KT [?], SINKT [?], Hyper-HKT [?], and GMGAE [?].